

Linear Regression

Concept

- A parametric model that fits a linear relationship between input features and a target variable.

Model

- Prediction formula:

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

Training Objective

- Minimize the mean squared error (MSE):

$$MSE = 1/m \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

Key Properties

- Fast and interpretable
 - Sensitive to outliers
 - Assumes linearity and independence
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Random Forest

Concept

- An ensemble of many decision trees built using bootstrap sampling and feature randomness.

How It Works

- Each tree is trained on a random subset of the data (bagging)
- Each split considers a random subset of features
- Final prediction:
 - Regression: average of all trees
 - Classification: majority vote

Strengths

- Handles nonlinear patterns
- Robust to outliers and noise
- Reduces overfitting compared to a single tree

Limitations

- Less interpretable
 - Can be computationally heavy
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XGBoost (Extreme Gradient Boosting)

Concept

- A high-performance implementation of gradient boosting that builds trees sequentially.
- Each new tree corrects the errors of the previous ensemble.

Objective Function

$$\mathcal{L} = \sum_{i=1}^m l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

Where

- l is the loss function (e.g., squared loss, log loss)

- Ω is a regularization term that penalizes tree complexity

Key Features

- Gradient-based optimization
- Regularization to prevent overfitting
- Handling of missing values
- Parallelized tree construction
- Highly efficient and accurate

Comparison Table

Model	Type	Strengths	Weaknesses
Linear Regression	Parametric	Simple, fast, interpretable	Assumes linearity, outlier-sensitive
Random Forest	Ensemble (Bagging)	Robust, handles nonlinear relationships	Slower, less interpretable
XGBoost	Ensemble (Boosting)	High accuracy, regularization, fast	Many hyperparameters